26900246 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFTQ303 - TECHNOLOGY OF SPICES

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Name two spices that possess antimicrobial properties. [K] / [CO1]
2. Name any two natural drying methods used for spices. [K] / [CO5]
3. List the major steps involved in the processing of nutmeg. [K] / [CO3]
4. Recall the gases used in fumigation of spices. [K] / [CO4]
5. Define cleaning of spices. [U] / [CO5]
6. List two applications of spice oleoresins in the food industry. [U] / [CO1]
7. Define ginger. [K] / [CO2]
8. What is black pepper? [K] / [CO2, CO3]
9. Differentiate between ethylene oxide treatment and conventional thermal treatment. [U] / [CO4]
10. Mention two methods of drying cardamom. [U] / [CO2, CO3]

11. State one important quality parameter of spice oleoresins. [U] / [CO1]

12. Name the major colouring principle present in turmeric. [K] / [CO5]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Describe the different methods of drying cardamom. [K] / [CO2, CO3]

14. Explain the major factors affecting the rate of drying of spices. [U] / [CO4]

15. Describe the volatility, aroma and flavour characteristics of spice oils. [U] / [CO1]

16. Differentiate between conventional extraction and supercritical carbon dioxide extraction of spice components. [U] / [CO1]

17. Differentiate between flower spices and bud spices. [U] / [CO5]

18. Differentiate between the processing methods of saffron and cloves, highlighting their major processing operations. [U] / [CO3]

19. Explain the advantages and limitations of irradiation treatment of spices. [U] / [CO4]

20. List the advantages of plastic packaging for spices. [K] / [CO5]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain the relationship between the chemical composition of spices and their functional properties in food preservation, flavouring and health promotion. [U] / [CO1]

22. Describe the various factors affecting the storage stability and shelf life of spices. [K] / [CO5]

23. Explain the mechanical methods of chilli drying in detail. [K] / [CO2, CO3]

24. Compare propylene oxide treatment with ethylene oxide and steam sterilization for spice decontamination. [U] / [CO4]

[10x2 = 20]